JEE ADVANCED – SAMPLE SYLLABUS

PHYSICS

Unit 1: Mechanics

- Motion in a Straight Line: displacement, velocity, acceleration
- Laws of Motion: Newton's laws, friction, circular motion
- Gravitation: universal law, orbital velocity, escape velocity

Unit 2: Electromagnetism

- Electrostatics: Coulomb's law, electric field, potential, capacitance
- Current Electricity: Ohm's law, Kirchhoff's laws, Wheatstone bridge
- Magnetic Effects of Current: Biot-Savart law, Ampere's law

Unit 3: Thermodynamics

- Laws of Thermodynamics, Carnot cycle, heat engines

CHEMISTRY

Unit 1: Physical Chemistry

- Atoms and Molecules: atomic structure, Bohr model
- Chemical Bonding: ionic, covalent, hydrogen bonding, VSEPR theory
- Chemical Equilibrium: Le Chatelier's principle, Kp, Kc

Unit 2: Organic Chemistry

- Carbon Compounds: hybridization, isomerism, reaction mechanisms
- Hydrocarbons: alkanes, alkenes, alkynes, aromatic compounds

MATHEMATICS

Unit 1: Algebra

- Matrices and Determinants: operations, inverse, Cramer's rule
- Sequences and Series: AP, GP, HP, sum of special series

Unit 2: Coordinate Geometry

- Straight Lines: slope, equation forms, angle between lines
- Circles, Parabola, Ellipse, Hyperbola

Unit 3: Calculus

- Integration: definite, indefinite, applications (area under curve)